

SUNDRE, AB, Canada

WMO: 712480

Lat: 51.778N

Lon: 114.683W

Elev: 1114

StdP: 88.63

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.2	-28.9	-37.2	0.1	-33.1	-32.6	0.2	-28.8	10.3	4.0	8.8	0.4	0.7	260	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	28.1	15.5	26.1	15.0	24.3	14.3	17.4	24.6	16.3	23.5	15.3	22.3	3.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	12.2	20.1	13.6	11.1	18.8	12.5	10.3	17.8	53.4	24.6	49.9	23.5	46.6	22.3	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.0	6.0	-38.4	31.7	4.0	2.0	-41.2	33.1	-43.5	34.3	-45.8	35.5	-48.6	36.9		
(5)			-38.3	19.2	3.8	1.2	-41.0	20.1	-43.2	20.8	-45.4	21.4	-48.1	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-9.7	-8.7	-3.9	2.6	8.0	12.0	15.2	14.1	9.4	3.3	-4.0	-9.5	(6)	
(7)		DBStd	10.70	8.54	7.72	7.71	4.80	3.99	2.86	2.86	3.07	3.99	4.61	7.12	7.97	(7)	
(8)		HDD10.0	3184	611	522	432	223	86	11	1	4	58	211	421	604	(8)	
(9)		HDD18.3	5801	869	756	691	471	319	189	103	135	268	466	671	863	(9)	
(10)		CDD10.0	434	0	0	0	2	25	73	163	130	39	3	0	0	(10)	
(11)		CDD18.3	9	0	0	0	0	0	1	6	2	0	0	0	0	(11)	
(12)		CDH23.3	596	0	0	0	1	31	49	241	204	69	2	0	0	(12)	
(13)		CDH26.7	128	0	0	0	0	3	8	56	47	13	0	0	0	(13)	
(14)	Wind	WSAvg	2.1	1.7	1.8	2.2	2.5	2.6	2.5	2.2	2.0	1.9	2.0	1.8	1.7	(14)	
(15)	Precipitation	PrecAvg	511	15	13	23	32	65	96	87	72	52	20	15	16	(15)	
(16)		PrecMax	761	28	33	47	120	153	232	196	135	120	41	30	41	(16)	
(17)		PrecMin	307	1	0	8	2	21	21	26	11	8	7	6	3	(17)	
(18)		PrecStd	111	7	8	10	26	30	60	42	32	31	11	7	12	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	11.7	15.4	21.8	26.6	27.8	30.5	30.6	28.5	22.7	15.6	11.2	(19)	
(20)			MCWB	3.5	4.0	6.1	9.5	11.9	14.9	16.7	15.5	14.0	10.5	6.8	4.1	(20)	
(21)		2%	DB	8.1	8.2	12.1	18.3	23.6	24.5	28.2	27.8	25.3	18.4	11.6	7.8	(21)	
(22)			MCWB	2.3	2.0	4.2	7.8	11.2	13.7	16.5	15.2	13.3	9.1	5.0	1.9	(22)	
(23)		5%	DB	5.7	5.7	9.6	15.4	21.0	22.5	26.0	25.6	22.6	15.3	8.7	5.0	(23)	
(24)			MCWB	0.8	0.5	2.9	6.3	10.3	12.9	16.1	15.1	12.5	7.4	3.0	0.2	(24)	
(25)		10%	DB	2.9	3.2	7.2	12.4	18.4	20.5	24.0	23.4	19.7	12.6	6.0	2.3	(25)	
(26)			MCWB	-0.8	-1.0	1.7	4.8	9.2	12.0	15.3	14.6	11.4	6.0	1.1	-1.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.2	6.5	9.9	13.4	16.7	19.6	18.2	15.6	11.6	7.3	4.3	(27)	
(28)			MCDB	9.8	10.8	14.2	19.6	22.5	24.3	26.1	25.5	25.1	21.2	14.2	10.6	(28)	
(29)		2%	WB	2.6	2.2	4.6	8.3	12.1	15.2	18.1	16.9	14.2	9.6	5.2	2.1	(29)	
(30)			MCDB	7.7	7.6	11.2	17.2	21.3	22.0	24.9	24.4	23.1	17.3	11.0	7.2	(30)	
(31)		5%	WB	1.1	0.7	3.1	6.7	10.9	13.9	16.9	15.9	13.0	8.0	3.4	0.4	(31)	
(32)			MCDB	5.5	5.3	9.1	14.5	19.3	20.4	24.1	23.1	21.5	14.5	8.2	4.5	(32)	
(33)		10%	WB	-0.7	-0.8	1.8	5.2	9.8	12.8	15.9	14.9	11.7	6.4	1.6	-1.3	(33)	
(34)			MCDB	3.0	3.0	6.8	11.6	17.3	19.0	22.6	22.1	19.3	11.8	5.4	2.1	(34)	
(35)	Mean Daily Temperature Range			MDBR	15.3	16.5	14.2	13.6	14.6	13.3	14.4	15.1	15.2	14.7	14.0	14.8	(35)
(36)		5% DB	MCDBR	17.3	19.5	18.1	20.1	20.9	18.6	19.1	20.8	22.4	20.1	17.2	16.7	(36)	
(37)			MCWBR	12.2	13.7	11.2	10.7	10.1	9.1	9.2	9.9	11.5	11.5	10.8	11.5	(37)	
(38)			MCDBR	16.9	18.9	17.1	18.8	18.6	15.7	16.6	17.7	20.5	19.1	16.6	16.1	(38)	
(39)		5% WB	MCWBR	12.1	13.4	10.7	10.3	9.4	8.2	8.8	9.1	10.9	11.2	10.6	11.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.254	0.262	0.290	0.335	0.344	0.352	0.347	0.361	0.316	0.280	0.255	0.241	(40)	
(41)		taud		2.410	2.468	2.450	2.337	2.375	2.380	2.408	2.376	2.466	2.562	2.500	2.390	(41)	
(42)		Ebn at Noon		786	886	917	903	908	901	900	866	874	854	787	741	(42)	
(43)		Edh at Noon		56	71	88	112	113	114	109	107	87	64	51	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg		1.09	2.03	3.34	4.55	5.35	5.65	6.17	5.12	3.62	2.23	1.19	0.84	(44)	
(45)		RadStd		0.06	0.14	0.20	0.34	0.42	0.45	0.41	0.35	0.42	0.22	0.08	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page